

Chapter 1

Introduction

Modern generative models of images, audio, language, and user behaviour increasingly share a common front end: a tokenizer that turns a continuous, high-dimensional signal into a short sequence of discrete symbols. This discretization is what lets a single autoregressive transformer or diffusion prior model heterogeneous modalities with the same machinery, and its quality sets a ceiling on everything downstream. The most useful tokenizers do more than compress; they impose *structure*. Residual quantization, in particular, decomposes a signal coarse-to-fine: the first code fixes the global gist, each later code refines the error left by its predecessors, and truncating the sequence at any depth yields a coherent, progressively sharper approximation [16, 7]. The tuple of codes is therefore not a flat label but a path from general to specific—a latent hierarchy. The promise of such tokenizers is that this hierarchy is learned for free, as a by-product of compression. The difficulty is that the geometry in which it is learned was never chosen to hold a hierarchy.

The discrete-bottleneck paradigm has been remarkably productive. The VQ-VAE [14] introduced a learnable codebook into the autoencoder and, with the straight-through estimator [2], made nearest-neighbour assignment trainable end to end; a decade of refinements has since tamed its optimization, from EMA updates and codebook resets to variational and scalar relaxations [13, 1, 8]. Residual vector quantization [16, 5] then lifted the idea to multiple stages, building a virtual codebook of exponential size K^N without enlarging the dictionary or the token sequence. The combination now underpins state-of-the-art neural audio codecs and the language-model audio systems built on them [3, 4], autoregressive image generators [7], and generative recommenders that assign items hierarchical semantic identifiers [11]. Across all of these, the same property recurs: the residual decomposition induces a tree-like organization of the data. If the representation is tree-like, the natural next question is whether the *space* that holds it is.

It is not. Every one of these methods quantizes in Euclidean space, whose volume grows only polynomially with radius. A tree’s node count grows exponentially with depth, so embedding a deep hierarchy into a flat space forces either crowding or distortion, and no choice of dimension removes the mismatch [12]. Hyperbolic space is the exact remedy: under constant negative curvature, volume grows exponentially, matching the branching of a tree, and any weighted tree embeds into the Poincaré disk with arbitrarily low distortion [9, 12]. The mismatch is sharpest precisely for residual quantizers, because the structure they induce *is* a tree, yet they are committed to the one geometry that represents trees worst. This makes hyperbolic space an unusually well-motivated home for residual codes, and motivates moving the quantizer onto the manifold.

Doing so faithfully, however, is more than a change of coordinates. Existing hyperbolic quantizers [10, 15, 6] place codes inside the ball but keep Euclidean training machinery—identity straight-through gradients, additive residual recursions transliterated into Möbius operations.

Our central observation is that this half-measure quietly breaks the two algebraic guarantees on which residual quantization depends, and breaks them for the same reason: the Poincaré ball’s addition is neither commutative nor associative, and its tangent spaces are rotated relative to one another by the gyration that encodes curvature. In the *forward* pass, the rule that removes a code from the residual is no longer inverted by the rule that re-assembles the codes, so the cascade stops telescoping: the tracked residual drifts from the true reconstruction error, deeper codebooks are fit to the wrong target, and the codes never recombine to the encoder point. In the *backward* pass, copying a gradient from a code to the encoder is not even type-correct—the two gradients live in different tangent spaces—and once stages are stacked, the exact cancellation that gives Euclidean RVQ a single clean gradient per step leaks, so reconstruction and commitment signals accumulate and explode with depth. Both pathologies are geometric in origin, so both admit geometric, rather than heuristic, repairs.

We introduce *Geometry-aware Hyperbolic Residual Quantization* (GHRQ-VAE), which treats quantization as a first-class operation on the Poincaré ball and repairs each pathology at its source. The forward repair, *Hyperbolic Residual Aggregation* (HRA), pairs a left-sided Möbius subtraction in the residual update with a reverse-nested aggregation, the unique pairing the gyrogroup’s left-cancellation law makes mutually inverse. This restores exact telescoping: the codes recombine to the encoder point, and the only remaining mismatch between tracked and true residual is a pure rotation that preserves magnitude, so each codebook again sees the genuine reconstruction error. The backward repair routes gradients at the *block* level: stop-gradients detach the intermediate residuals, and a single *discounted hyperbolic straight-through estimator* transports one geometrically correct, depth-independent gradient from the aggregate reconstruction directly to the encoder, reproducing the Euclidean “firewall” on the manifold while a cancellation-free gyration keeps it bounded near the boundary, where curvature is most expressive. The construction is curvature-faithful by design: at $c = 0$ the Möbius operations collapse to vector addition, the gyration to the identity, and GHRQ-VAE recovers standard residual vector quantization exactly. Hyperbolic quantization thus becomes a strict generalization of its Euclidean counterpart, not a separate model.

We evaluate GHRQ-VAE against a Euclidean baseline and the naive hyperbolic lift across four modalities chosen to span the regimes where these methods are used and where they break: WordNet hypernymy prediction, generative sequential recommendation, image reconstruction and generation on MNIST and CIFAR-100, and neural audio coding on LibriTTS—the last at a depth ($n_q = 12$) far beyond what prior hyperbolic residual quantizers have attempted, and the first application of hyperbolic residual quantization to a convolutional image tokenizer and a deep audio codec. The geometry-aware repairs deliver what they were designed to: the on-ball residual error falls by one to three orders of magnitude relative to the naive lift, confirming that the cascade telescopes in practice; codebook collapse is eliminated where the naive lift suffers it; and the learned codes recover latent taxonomies markedly better, raising unsupervised CIFAR-100 superclass agreement by 80% in Adjusted Rand Index and tightening WordNet semantic clusters. Component ablations localize these gains, showing that the corrected gradient—not merely the forward ordering—is what organizes hierarchy. We also report an honest negative result: Euclidean quantization remains the better pure compressor on raw reconstruction and perceptual fidelity. Curvature does not buy distortion; it buys structure and stability, and it buys them robustly with depth.

In summary, our contributions are:

- **A precise diagnosis.** We show that naively lifting residual quantization to hyperbolic space breaks its two defining guarantees, and identify the geometric cause of each: a forward cascade that no longer telescopes because Möbius addition is non-associative, and a backward “firewall” that leaks because the per-stage gradient is neither transported nor cancelled correctly, causing gradients to accumulate with depth.

- **A geometry-aware quantizer.** We propose GHRQ-VAE, combining Hyperbolic Residual Aggregation, which restores exact telescoping up to a norm-preserving rotation, with block-level gradient routing through a numerically stabilized discounted hyperbolic straight-through estimator, which delivers a single depth-independent gradient to the encoder. The formulation reduces to standard Euclidean RVQ at zero curvature.
- **Evidence across four modalities.** We provide the first study of hyperbolic residual quantization on image and deep-audio tokenizers, demonstrating near-exact on-ball reconstruction, restored codebook health, and improved unsupervised hierarchy recovery, together with ablations isolating each component and a clear account of where curvature helps (structure, stability) and where it does not (raw fidelity).

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